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DATE:18.03.26

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EXPERIMENT 8

PERFORM ROOTKIT DETECTION AND REMOVAL USING RKHUNTER TOOL

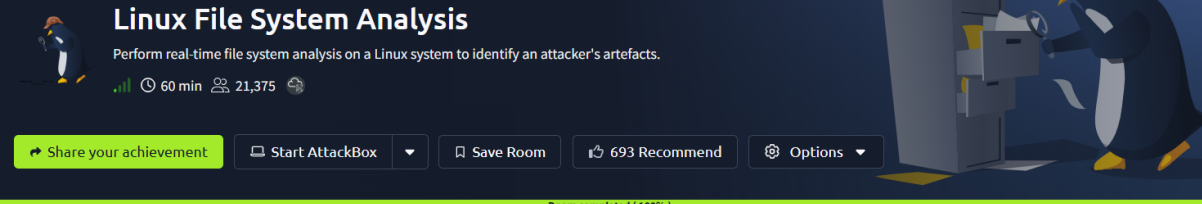
AIM:

To analyze and understand the Linux file system structure, permissions, and file management techniques, and to identify important system files using the TryHackMe lab environment.

ALGORITHM:

1. Log in to the TryHackMe platform and start the Linux File System Analysis room.Observe the network diagram and devices.
2. Navigate directories using commands like ls, cd, and study key directories such as /home, /etc, /var, and /root.
3. Use commands like ls -l, chmod, and chown to understand read, write, and execute permissions.
4. Locate important or suspicious files using commands like find, grep, and cat.
5. Interpret results to understand system organization, user access, and identify critical or sensitive files.

TASKS:



Linux File System Analysis

Perform real-time file system analysis on a Linux system to identify an attacker's artefacts.

📶 60 min 👤 21,375 🌐

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Room completed (100%)

Task 1 Introduction

Task 2 Investigation Setup

Task 3 Files, Permissions, and Timestamps

Task 4 Users and Groups

Task 5 User Directories and Files

Task 6 Binaries and Executables

Task 7 Rootkits

Task 8 Conclusion

Answer the questions below

After updating the `PATH` and `LD_LIBRARY_PATH` environment variables, run the command `check-env`. What is the flag that is returned in the output?

THM{5514ec4f1ce82f63867806d3cd95dbd8}

✓ Correct Answer



Answer the questions below

To practice your skills with the `find` command, locate all the files that the user **bob** created in the past 1 minute. Once found, review its contents. What is the flag you receive?

THM{0b1313afd2136ca0faafb2daa2b430f3}

✓ Correct Answer



Extract the metadata from the `reverse.elf` file. What is the file's MIME type?

application/octet-stream

✓ Correct Answer

Run the `stat` command against the `/etc/hosts` file on the compromised web server. What is the full **Modify Timestamp (mtime)** value?

2020-10-26 21:10:44.000000000 +0000

✓ Correct Answer

Answer the questions below

Investigate the user accounts on the system. What is the name of the backdoor account that the attacker created?

b4ckd00r3d

✓ Correct Answer



What is the name of the group with the group ID of **46**?

plugdev

✓ Correct Answer

View the `/etc/sudoers` file on the compromised system. What is the full path of the binary that Jane can run as sudo?

/usr/bin/pstree

✓ Correct Answer

Answer the questions below

View Jane's `.bash_history` file. What flag do you see in the output?

THM{f38279ab9c6af1215815e5f7bbad891b}

✓ Correct Answer

What is the hidden flag in Bob's home directory?

THM{6ed90e00e4fb7945bead8cd59e9fcd7f}

✓ Correct Answer

Run the `stat` command on Jane's `authorized_keys` file. What is the full timestamp of the most recent modification?

2024-02-13 00:34:16.005897449 +0000

✓ Correct Answer

Answer the questions below

Run the `debsums` utility on the compromised host to check only configuration files. Which file came back as altered?

/etc/sudoers

✓ Correct Answer

What is the `md5sum` of the binary that the attacker created to escalate privileges to root?

7063c3930affe123baecd3b340f1ad2c

✓ Correct Answer

Answer the questions below

Run `chkrootkit` on the affected system. What is the full path of the `.sh` file that was detected?

/var/tmp/findme.sh

✓ Correct Answer

Run `rkhunter` on the affected system. What is the result of the `(UID 0) accounts` check?

Warning

✓ Correct Answer

Answer the questions below

Click and continue learning!

No answer needed

✓ Correct Answer

RESULT:

All the tasks in Linux File System Analysis tryhackme room have been completed successfully.